A Market World Model for Public LLMs: Complete, Honest, Auditable Crypto Cognition before Decision

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Abstract

Most quant stacks start from *strategy*. Public large language models (GPT-, DeepSeek-, and GLM-class agents) that act in cryptocurrency markets instead fail earlier: they lack a complete, honest, auditable *market cognition base*. Raw feeds are asynchronous, intermittently missing, and quality-heterogeneous; ungated tool dumps invite overconfident action. We build a *financial world-model runtime* that compiles multi-band evidence into a point-in-time (PIT) world bundle for public LLMs. The primitive is an epistemic observation $O_{j,t} = (x, \tau, q, g, r)$; a compilation operator Π_t yields $\mathcal{F}_t^{\text{AI}}$; completeness, honesty (B, U, H), and WMI/ACWMI gate abstention; evidence IDs make actions auditable. On a real crypto PIT panel, Compiled-versus-Raw consumers that honor the world model abstain on thin worlds (thin-world abstain rate 1.0, EAR proxy 1.0), while momentum-only and noisy consumers do not. A secondary content probe shows vintaged macro/stablecoin fields change a transparent rule versus same-input momentum (ΔCE 0.334, p=0.034)—evidence the compiled world is nonempty, not a strategy Holy Grail. Live OpenAI-compatible adapters (GPT / DeepSeek / GLM) share the same protocol when API keys are present.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics.*

Keywords

world models, public LLMs, AI agents, cryptocurrency, point-in-time, abstention, trustworthy AI, market cognition

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1 Introduction

Public LLMs are increasingly asked to analyze or trade crypto markets. The bottleneck is not another alpha signal. It is *understanding*: before decision, an agent needs a market world that is **complete** (missingness disclosed), **honest** (stale or role-incoherent evidence gated), and **auditable** (actions bound to evidence). Venue fragmentation, derivatives, on-chain flows, news, and macro vintages arrive on incompatible clocks [9, 10]. Generative world models [5, 6, 8] and tool-using agents [1, 13] typically assume a usable observation stream. Crypto agents often do not have one.

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Thesis. We build a market *world-model runtime* for public LLMs: compile raw multi-band evidence into a PIT-safe cognition bundle that GLM / DeepSeek / GPT-class models can call. Strategy metrics are secondary probes that the world has content—not the product objective.

Contributions.

(1) **Cognition semantics.** Epistemic observations, compilation Π_t, reconstruction bound, completeness/honesty fields, WMI/ACWMI abstention, evidence-bound actions.
(2) **Runtime for public LLMs.** Vintage-aware collectors, Band-PIT previous-close clock, quality-tagged bundles, availability shocks O_t, frozen Compiled-versus-Raw protocol with OpenAI-compatible live adapters.
(3) **Understanding-first validation.** On a real PIT archive, world-honoring consumers abstain when the world is thin; content probes show compiled band fields are economically nonempty versus same-input momentum.

2 Related Work

World models. World models learn compact states and often dynamics [5, 6, 8]. We address the complementary *observation-side* problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy.

LLM agents in finance. Agentic workflows call tools over streams [13]. Without a compiled cognition base, models face stale or incomplete evidence. We evaluate within-model Compiled vs. Raw under a frozen schema.

Selective prediction and trustworthy AI.. Abstention can be optimal under noise [3, 4]. We tie refusal to world quality, not classifier confidence alone—aligned with trustworthy AI in finance.

PIT measurement. Macro vintages [2] and crypto microstructure [9, 10] motivate multi-band compilation. Bootstrap discipline follows [7, 11, 12].

3 Market World-Model Semantics

Definition 1 (Epistemic observation). *$O_{j,t} = (x_{j,t}, \tau_{j,t}, q_{j,t}, g_{j,t}, r_{j,t})$: value, latest-available time, quality, main-view gate, semantic role.*

Definition 2 (Compilation). *$W_t = \Pi_t(\mathcal{F}_t^{\text{raw}})$ with $\Pi_t = B_t \circ M_t \circ A_t$ (align, honesty/missingness gates, band assembly); $\mathcal{F}_t^{\text{AI}} = \sigma(W_t)$.*

Completeness, honesty, quality. Ready/limited/missing counts disclose coverage. Breadth/stability/honesty give $\text{WMI}_t = B_t U_t H_t$; ACWMI supports regime-conditional gating. Reconstruction error admits a delay/noise/missingness bound (Eq. omitted for space).

Proposition 1 (Compilation ≠ feature dump). *Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{\text{AI}}$: ungated stale evidence can raise raw volume while lowering honesty/stability.*

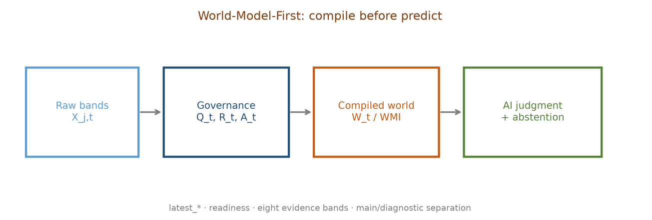

Figure 1: World-model runtime: raw crypto evidence → compiled cognition bundle → public LLM decision / abstain.

Table 1: World bundle served to public LLMs (cognition base).

Group	Fields
Timing	`decision_asof`, previous-close protocol
Completeness	$n_{\text{ready/limited/missing}}$, band statuses
Honesty	B, U, H, main-view / stale gates
Quality	WMI, ACWMI, `should_ai_abstain`
Content	macro_tilt, alt_tilt, regime, cascade
Audit	`evidence_ids`, EAR required

Proposition 2 (World-conditional abstention). *If non-abstain actions exceed a world-dependent cost $c_{\text{abs}}(W_t)$, Bayes action is abstain; with costs decreasing in WMI, abstention is a lower set in world quality [3].*

Remark 1 (Not a generative WM). *We do not learn $p(s_{t+1} \mid s_t, a_t)$. We deliver a* state compiler + abstention runtime *for public LLM consumers.*

4 Runtime for Public LLMs

Figure 1 shows the anonymized prototype. Collectors fill vintage-aware stores; BandPIT reconstructs readiness at $(t-1)$ 23:59 for payoff r_t; the runtime emits a world bundle consumed by rule or LLM agents via frozen prompts (temperature 0). OpenAI-compatible adapters target GPT / DeepSeek / GLM when API keys exist; offline stylized public-LLM followers keep the protocol reproducible without keys.

Compiled vs. Raw protocol.

$$\Delta_m = V_m(\text{Compiled}) - V_m(\text{Raw}),$$

same dates/assets/schema. Understanding metrics: thin-world abstain rate; EAR proxy; disclosed ready share. Durable bands in the archive: {exchange, macro, alternative} (ready rates ≈ 0.80/1/1); other bands right-censored and disclosed as missing.

5 Experiments

5.1 Setup

PIT panel ~399 days, 10 liquid names, previous-close clock, IS/OOS 200/200 days (~2000 asset-days OOS). Consumers: `public-llm-compiled-follower` (stylized public LLM), diagnostic mocks (compiled-aware / momentum / noisy), optional live vendors. Prompts frozen; temperature 0.

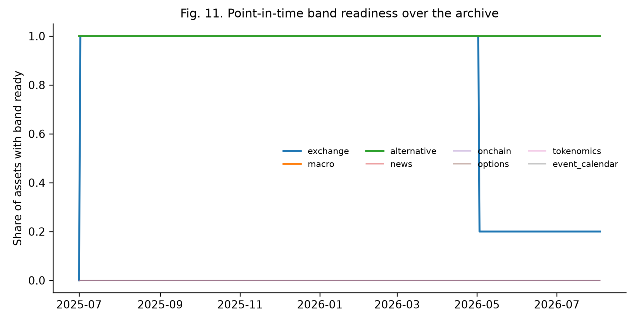

Figure 2: PIT readiness: completeness is partial—missing bands are first-class in the bundle.

Table 2: Compiled vs. Raw public-LLM protocol (OOS). CE is secondary.

Consumer	ΔCE	Abs. C	Thin-abs.	EAR
public-llm-follower	+0.202	1.00	1.00	1.00
mock-compiled-aware	+0.202	1.00	1.00	1.00
mock-momentum	0.000	0.00	0.00	1.00
mock-noisy	+0.212	0.00	0.00	1.00

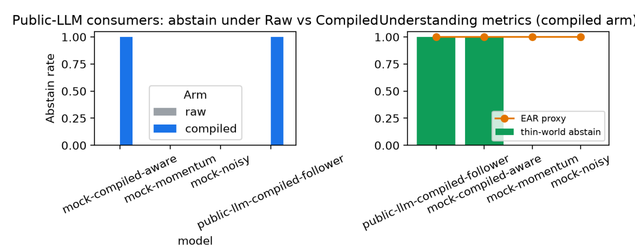

Figure 3: Understanding-first metrics: compiled arms refuse thin worlds; raw/momentum/noisy do not.

5.2 RQ1: Does the world model change LLM behavior?

Table 2 and Figure 3: world-honoring consumers abstain on the entire OOS window under the sparse-archive WMI/ACWMI gate (abstain 1.0; thin-world abstain 1.0; EAR proxy 1.0). Raw arms never receive abstention guidance (abstain 0). Momentum-only and noisy consumers *ignore* thin-world flags (thin-world abstain 0)—exactly the failure mode of ungated public LLMs. Mean ΔCE across consumers is positive but secondary: the cognition claim is correct refusal under thin support (Prop. 2).

Table 3: Secondary content probe (not the LLM product claim).

Policy / contrast	Sharpe	CE
Momentum always	0.101	−0.202
Thick ungated (content)	0.767	0.132
Mechanism − Momentum	—	0.334 (p=0.034)

5.3 RQ2: Is the compiled world nonempty? (content probe)

To show the bundle is not an empty label, a transparent rule shares return inputs with momentum and adds vintaged macro_tilt / alt_tilt. Pre-specified OOS contrast mechanism − momentum: ΔCE=0.334 (p=0.034; stationary p=0.022). LOBO: deleting macro or alternative collapses to momentum (content share 1.0 under the ungated rule). Relative gaps survive 10–25bps costs; absolute CE is cost-fragile. A 2017–2026 audit without vintaged archives finds no hidden return-rule alpha. These results *probe content*, they do not define the product.

5.4 Discussion

Quant stacks usually ask "what trades?" We ask "what does the LLM know, and should it act?" Table 2 answers the second question under sparse support: a cognition base that marks the world thin induces abstention; raw feeds and strategy-like mocks do not. Table 3 answers a weaker but necessary question: compiled band fields are not decorative—they change actions and CE versus same-input momentum. Together they match the product claim: *understanding first, strategy second.*

Live vendors share the OpenAI-compatible adapter without changing the estimand. In this environment, API keys were absent, so we report the offline public-LLM follower and diagnostic mocks; the same prompts and bundles apply when GPT / DeepSeek / GLM endpoints are configured.

6 Limitations

Offline stylized public-LLM followers are protocol-complete without keys; live GPT/DeepSeek/GLM runs require credentials and remain the next measurement step. Sparse archives make production WMI thresholds bind almost always—consistent with support-matched abstention (Prop. 2), not a finished selective-prediction horse-race on dense worlds. News/on-chain/options history remains right-censored and is disclosed as missing. We do not claim generative dynamics $p(s_{t+1} \mid s_t, a_t)$ or unconditional trading alpha.

7 Conclusion

Public LLMs need a market to *understand* before they decide. We compile asynchronous crypto evidence into a complete, honest, auditable world-model bundle and show that world-honoring consumers abstain when the world is thin, while content probes confirm the compiled state is nonempty. Strategy is downstream of understanding—not the other way around.

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